Defining Core Routes in routes.py – FitBuddy

Core Routes

Separate routes were implemented to serve each of FitBuddy's main functionalities:

• / : Displays the user form via index.html

• /generate-workout : Processes user input and generates a personalized workout plan

• /submit-feedback : Updates the workout plan based on user feedback

• /view-all-users : Displays an admin dashboard showing all users and their plans

Set Up Route Handlers for Each Feature

➤ / – Home Route

• Returns the homepage (index.html)

• No form processing logic

➤ /generate-workout – Plan Generator

• Receives user input using FastAPI's Form(...)

• Constructs a UserInput Pydantic model

• Calls generate_workout_gemini(...) to generate a workout plan via Gemini 1.5 Pro

• Calls generate_nutrition_tip_with_flash(...) to get a nutrition tip via Gemini Flash

• Stores user details with save_user(...)

• Stores the plan with save_plan(...)

• Returns result.html with dynamic data

➤ /submit-feedback – Update Plan with Feedback

• Captures feedback and user_id

• Retrieves the original plan from the database

• Uses update_workout_plan(...) to revise the plan via Gemini 1.5 Pro

• Updates the revised plan in the database using update_plan(...)

• Renders the updated result on result.html

➤ /view-all-users – Admin View

• Calls get_all_users() and get_all_plans() to fetch database records

• Passes them to all_users.html

• Admin can view: Username, Age, Weight, Goal, Intensity

• Admin can view original and updated workout plans

Integrate Gemini AI Responses in Each Function

Each AI-related route calls Google's Gemini models with specific prompts:

• **Workout Plan Generation:** generate_workout_gemini() sends fitness goal and intensity to Gemini Pro to return a structured 7-day workout plan.

• **Nutrition Tip Generation:** generate_nutrition_tip_with_flash() uses Gemini Flash to generate a practical tip related to the user's fitness goal.

• **Workout Plan Update via Feedback:** update_workout_plan() passes the original plan plus user feedback to Gemini Pro and returns a revised plan.

All responses are returned to the user through result.html, maintaining clarity using blocks for formatting.

Route / Feature Summary

• Workout generation route

• Nutrition/Recovery tip generation route

• Saving user information and plan generation route

• Submit Feedback and update plan route